- Predict the next Token

The | cat | sat | on | the | ?

AI predicted :

mat (45%)
floor (28%)
roof
wall
...

- Discriminative

$y = f(x)$; seperating hyperplain

- Generative

$x_t = g(\underbrace{x_{<t}}_{\text{ใส่ข้อมูลก่อนหน้า}})$; autoregressive

- Paper : Attention is all you need (2018)

1. Pre-training : Predict the next token on
2. Instruction Tuning : Supervised fine-tuning (SFT)

   Generative : $t_1, t_2, t_3,$ [next] t_4?

   SFT : keyword (prompt) : before grammar : t_1, t_2, t_3
   after grammar : t'_1, t'_2, t'_3

3. Alignment : Reinforcement Learning from Human Feedback (RLHF)

   good response
   bad response } Human based

อาจจะมั่ว
ได้!
(ทำแค่ 3 ขั้นตอน)

(Hallucination)

Generative : $t_1, t_2, t_3,$ [next] t_4?

SFT : keyword (prompt) : before grammar : t_1, t_2, t_3

after grammar : t'_1, t'_2, t'_3

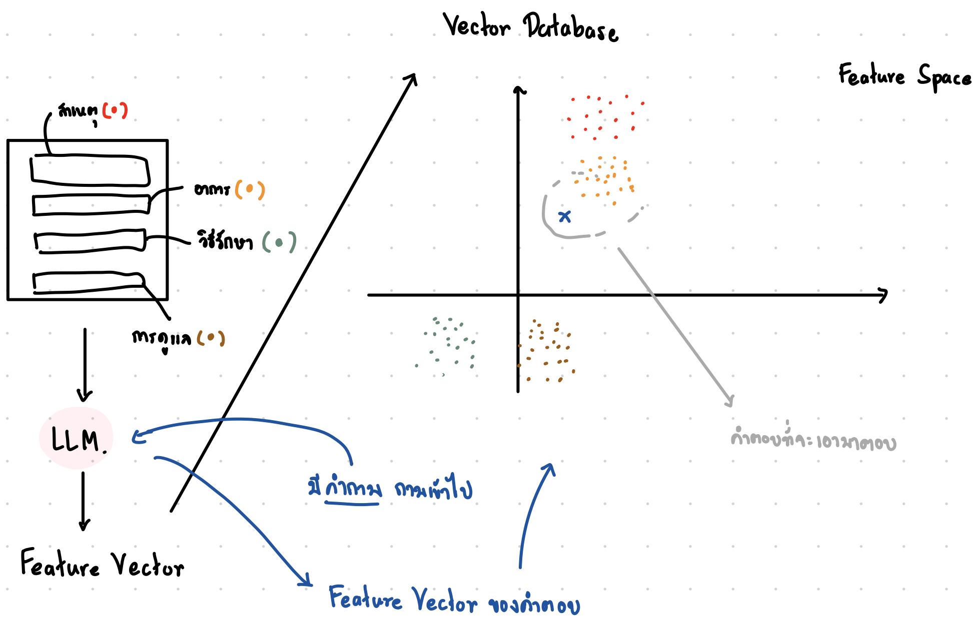
(Hallucination)

Retrieval-Augmented Generation (RAG)

context : C_1, C_2 $\longrightarrow$ Vector Database

before : t_1, t_2, t_3

after : t'_1, t'_2, t'_3



Generative : $t_1, t_2, t_3,$ [next] t_4 ?

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(Hallucination)

Retrieval - Augmented Generation (RAG)

context : $C_1, C_2 \longrightarrow$ Vector Database

before : t_1, t_2, t_3

after : t'_1, t'_2, t'_3

Agentic

run R code, Edited R code

$C_1, Q_1,$ $Tool_1$, $Tool_2$, $Tool_3$, $Tool_4$, Ans.

gen R code, Debug R code